Sravanth Kumar Raju Konduru

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.54
Narayana Junior college	Class XII (Board of Intermediate Education, Andhra Pradesh, India)	2021	92.00%
Raju High School	Class X (Board of Secondary Education, Andhra Pradesh , India)	2019	9.70%

SKILLS

- **Developer Tools:** Matlab/ Simulink , PLECS, Homer pro, Latex
- **Programming Skills:** C , Python, PyTorch, Tensorflow
- **Developer Kits:** Arudino IDE, C2000

RELEVANT COURSES

Power Systems Engineering and Converters for Renewable Applications| Control of Power Converters for Solar Photovoltaic|

Photovoltaic (PV) Technology| Electric Vehicles| Smart Grid| Electrochemical Energy Storage Systems

PROJECTS

M.Tech Thesis - Edge AI based improved Efficiency and Decoupled control of Triple Active Bridge (TAB) converter for Renewable battery charging Infrastructure. (Jul 2026 - Present)

- Developed a mathematical model and decoupled control strategy for a Triple Active Bridge (TAB) converter, enabling independent power control across ports while reducing circulating current and cross-coupling effects.
- Designing an Edge-AI-based adaptive control framework that adjusts phase-shift parameters in real time, improving efficiency and stability against renewable source variations and charging load disturbances.
- To Built and test a TAB converter prototype on a DSP platform, validating the proposed Edge-AI controller's performance against conventional control methods for renewable-based DC charging applications.

Feed-Forward based Decoupled Control of PV-Assisted TAB for EV Charger. (May - Jun 2026)

- Developed a TAB Converter for a PV-assisted EV charging Infrastructure, Identifying coupled power flow between ports.
- Feed forward based decoupling control is designed to achieve Independent voltage control across the ports.
- Designed an PI based SPS Control of EV Charging as well PV MPP for various loading and Environmental conditions.
- Performed simulation and scale down Hardware for the Proposed system.

A Two-Way approach to Achieve Net-Zero Buildings (Jan - may 2025)

- Conducted an energy audit at NIT Puducherry, identifying outdated equipment, wasted energy and untapped renewable integration and Proposed a practical retrofit solutions to reduce the Load consumption and carbon Emissions.
- Designed a fully Renewable hybrid system and evaluated by analysing the energy savings using HOMER Pro Software.
- Techno-economic Analysis is done to determine the cost effective and most reliable renewable energy system using HOMER pro software.
- Analyzed the Carbon reduction and economic benefits of the proposed System towards sustainability and net-zero building Strategy.

B.Tech Thesis - Performance Analysis and Aging Study of Transformer Oil Samples for High Voltage Applications (Jul - Dec 2024)

- Conducted BDV, viscosity, flash point, and fire point tests on transformer oil samples collected from substation.
- To study the aging effects and degradation patterns Synthetic ester is blended with fresh mineral oil and Aged transformer mineral oil to improve transformer reliability.
- Analysed results showing improved dielectric strength, viscosity, and flash/fire points, for enhanced transformer oil performance and longevity.

PUBLICATIONS

1. K. Sravanth Kumar Raju, Pathloth Pavan, Rajvir Kaur, Saurabh Kumar, "A Two-Stage Approach to Achieve Net Zero Energy Buildings: A Case Study," Sustainable Energy Technologies and Assessments, vol.91, p.105051, 2026. DOI: 10.1016/j.seta.2026.105051.
2. Sirisha Devi Ponnaganti, Praveen Vankadari, Konduru Sravanth Kumar Raju, Rupesh Wandhare, Sankar Peddapati, "Feed-Forward based Decoupled Control of Solar-Assisted Triple Active Bridge based Electric Vehicle Charger," IEEE PEDES(2026). (Accepted for Presentation)

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant** -- for the course, EE3501, Power Electronics. **(Jul26 - present)**
- **Placement Coordinator** -- EST, IIT Hyderabad **(May26 - present)**
- **Event Coordinator** -- SPREE, NIT Puducherry **(2022 - 2023)**
- **Volunteer** -- Gyanith Tech Fest, NIT Puducherry **(2022 - 2023)**

EXTRACURRICULAR

- Active player of Badminton.
- Represented the EEE department in the Inter Departmental Kho-Kho Team.